Introduction and Demonstration of All-Weather Portfolio

谭力滔





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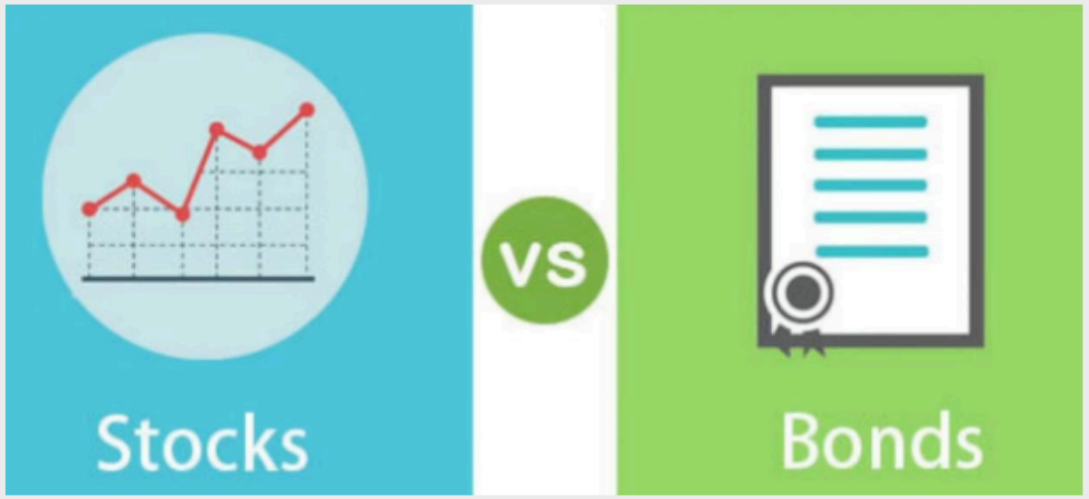
03 Instructions and references



•What is a portfolio

什么是资产配置





Traditional portfolio

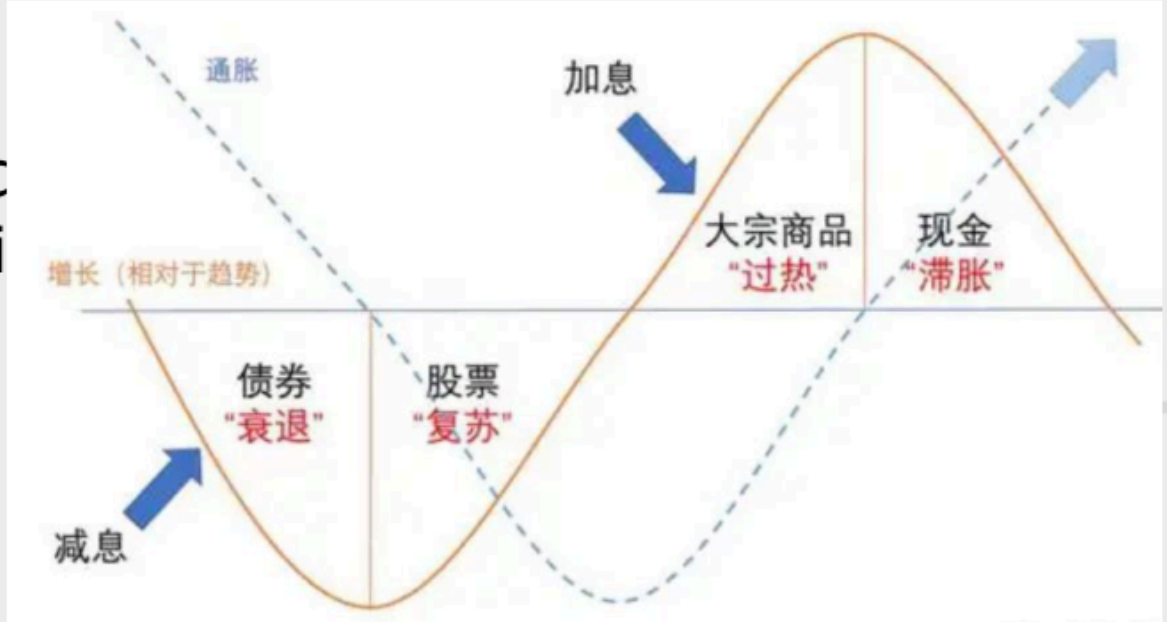
both stocks and bonds are allocated

(Stocks and bonds are usually negatively correlated)

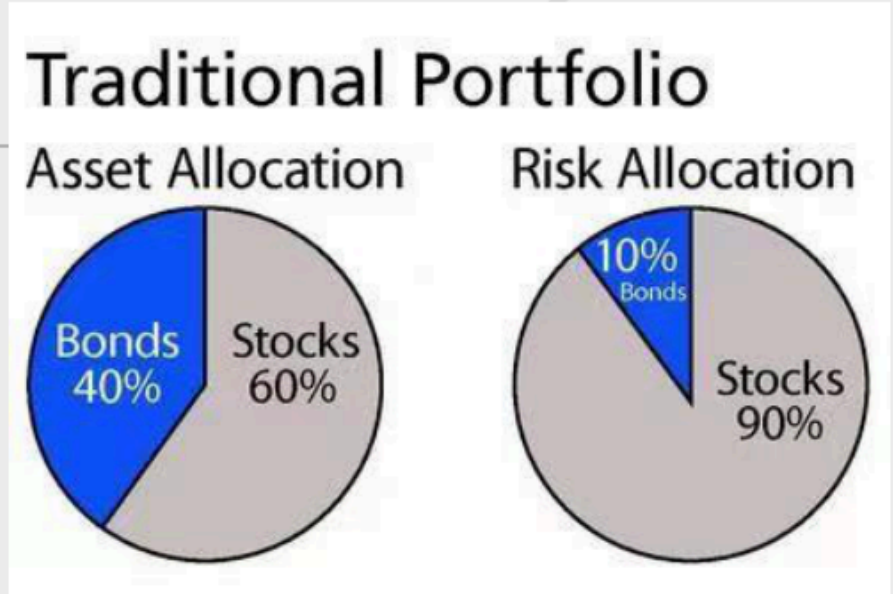
•Disadvantages of traditional portfolios

传统资产组合的缺点

stocks and bonc
have different ri



Difficult to predict the economic environment



All-Weather Portfolio

全天候策略资产配置



•Four grid model:

The business cycle is divided into four economic environments, and each environment has a corresponding asset subportfolio.

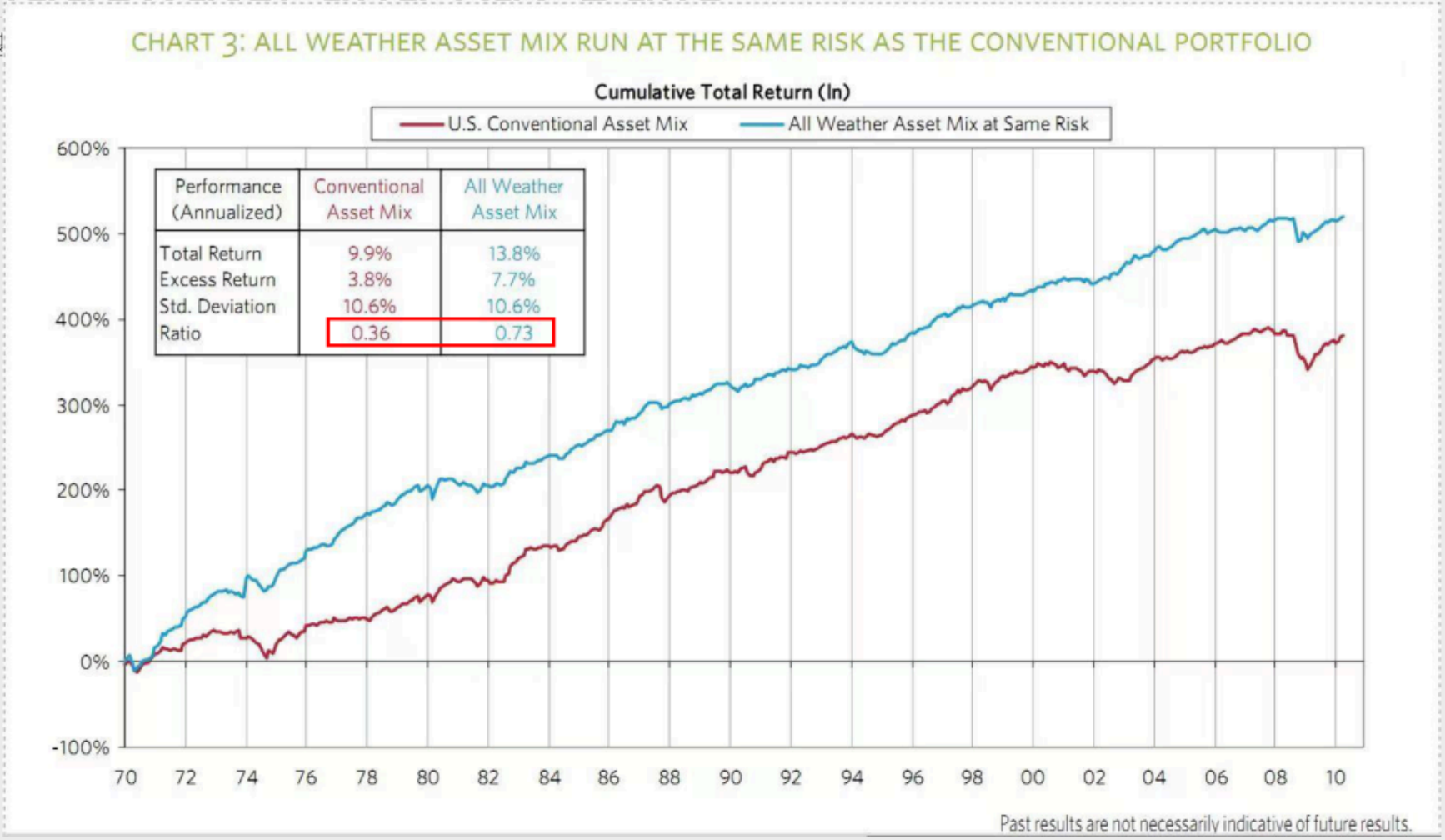
•Risk Parity:

All four economic environments are equally likely to occur, requiring only a 25% risk asset allocation in each environment.

Advantages: No matter how the economic environment changes, the volatility of the portfolio is very small, can guarantee stable returns

Bridgewater and Ray Dalio

桥水基



The Sharpe Ratio of AW vs traditional portfolio with the same fluctuation (60:40) is 0.73 and 0.36, respectively. (The higher the Sharpe ratio, the better the fund's return risk performance)

Correlation analysis

相关性分析

B	C	D	E	F
600519.SH	000001.SZ	601166.SH	511260.SH	600690.SH
0.57	2.08	1.29	0.0315135	0.86
-1.14	-1.37	-2.08	-0.055330921	-0.23
0.88	1.60	0.61	0.035765075	0.63
-0.30	-1.71	-0.76	0.084374521	1.59
0.10	0.27	-1.10	-0.09791484	-1.80
0.81	0.27	1.94	0.168868496	1.51
-2.29	-2.74	-1.70	0.108432089	-1.21
-2.40	-2.35	-2.16	-0.042671582	-3.55
0.47	0.00	-0.39	-0.025596396	2.24
-0.36	-2.11	-2.71	0.107620561	5.63
2.23	0.00	2.48	-0.125399872	3.11
3.60	6.25	0.59	-0.068198287	0.34
1.43	-0.81	0.69	0.087028489	-0.25
-3.67	-7.53	-5.32	0.103346373	-3.37
-0.39	1.58	1.37	-0.017079128	2.33
-0.95	-0.25	-0.61	0.16594245	-1.76
0.37	0.25	-0.89	-0.145202344	2.34
-0.03	-0.57	-1.06	-0.08448757	-1.12
-0.84	-3.17	-2.61	-0.156781214	-1.91
-0.17	2.37	2.68	-0.126797719	-0.80
2.26	1.52	-0.18	0.118427891	4.11
-0.73	-0.75	-0.05	-0.078322536	1.83
3.60	-0.81	-0.09	0.052810453	3.95
-3.59	-2.13	-1.09	-0.067245489	-1.23
1.08	0.74	1.85	0.202144264	-0.74
0.00	-0.67	-1.51	0.058886784	-2.62
-0.84	4.06	2.43	0.091400651	-0.55
3.55	2.41	3.53	0.15827929	2.81
-0.64	1.12	1.87	-0.132436751	3.13
3.79	3.61	2.06	0.065834473	4.43
0.37	-1.14	-0.75	0.035922612	-1.56
-1.73	-0.87	-0.20	0.027377102	-0.50
-1.80	-1.45	1.11	0.162816206	-2.19
-1.78	1.33	0.71	0.024857072	0.58
3.37	-1.19	-5.29	0.034297375	-0.54
-0.55	4.04	2.92	-0.049706475	0.63
-0.20	-0.75	-2.60	-0.122402164	0.18
-0.72	1.73	4.27	0.152593228	-0.13

	90.SH
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60116	
51126	
60069	1

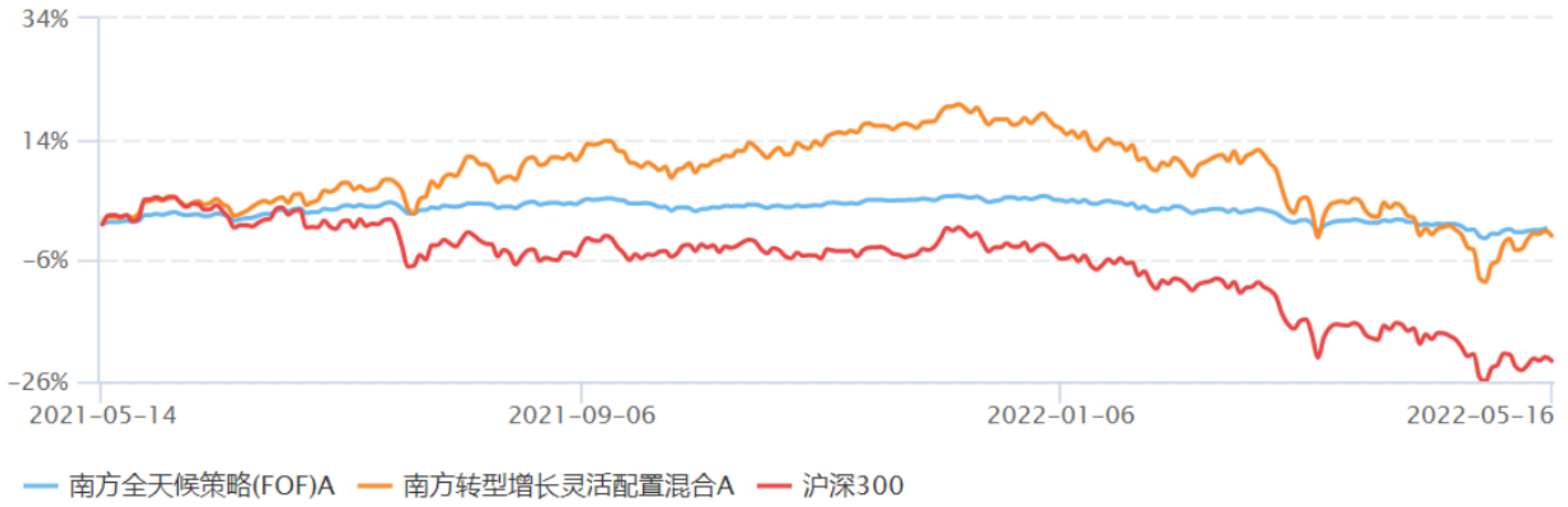
H	I	J	K	L
006491.OF	040026.OF	002100.SZ	603668.SH	002648.SZ
0.035778175	0.0356157	-0.54	0.09	1.83
0.035790981	0.0356284	3.03	-1.13	0.53
0.008948546	0.0089079	0.11	-0.52	2.34
0.026852846	0.0356443	0.56	0.61	0.17
0	0	-2.21	1.59	5.05
0.026860059	0.0267404	0.00	2.35	-2.61
0.053748992	0.0624331	4.14	1.00	-3.56
0	0	-2.36	-4.02	-2.67
0.044810898	0.044615	-0.67	3.07	0.92
0.02689377	0.0267762	-0.99	2.97	4.89
-0.035845506	-0.0356888	-3.93	0.09	0.03
-0.026876904	-0.0267594	4.09	2.67	4.19
0	0	-6.31	-4.73	-1.99
0.053782718	0.0535475	-9.38	-6.78	-6.56
0	0	1.52	2.97	-2.57
0.026898592	0.0357111	-7.81	-4.66	-6.15
0	0	-3.06	-2.91	-1.99
0.02690583	0.0267905	-2.25	0.65	0.58
-0.035861574	-0.0446309	5.90	1.07	0.20
0.035874439	0.0446508	-2.74	-1.62	-2.33
0.062819707	0.0714924	-0.26	-0.80	0.02
0.035909866	0.035759	-2.58	-1.34	1.19
0	0	2.83	2.76	-2.25
-0.01795171	-0.0178763	2.10	-4.06	2.13
0	0	-5.06	-0.47	4.64
0.017954933	0.0178795	-5.34	-4.31	-2.70
0.07187135	0.0805225	2.00	1.89	-1.81
0.062927005	0.0626679	2.55	0.92	2.54
0.026975991	0.035823	-3.26	-2.68	-2.11

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Back-test

业绩回测

业绩走势对比



Instructions

相关说明

1. The success of Bridgewater Fund is not only due to the all-weather strategy, but also due to other factors.

2. The content of all-weather portfolio is actually more complicated.

3. The all-weather strategy is not perfect, and there have been blowouts and underperformance.

4. This class presentation does not involve any investment advice.

Reference

引用

https://www.zhihu.com/answer/765993675

https://zhuanlan.zhihu.com/p/25105939

https://b23.tv/AodheBf

